

MARK YANG

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EDUCATION

Northeastern University

Boston, MA (December 2027)

- BS in Computer Engineering (3.98 GPA)
- Relevant Coursework: Embedded Design, Circuits and Signals, Fundamentals of Networks, Neuromechanical Simulation of Human Movement, Computer Vision, Engineering Algorithms, Linear Systems
- Activities: AI Club (Research Lead), Tau Beta Pi Engineering Honors Society, Putnam Club

PROFESSIONAL EXPERIENCE

NEUMove

Undergraduate Researcher

Boston, MA (Aug 2025–Present)

- Programmed the compliant muscle tendon unit (CMTU) in MuJoCo alongside a PhD collaborator and finished fitting the CMTU to existing OpenSim models for integration with the lab's direct collocation repository.
- Developing GPU-accelerated reinforcement learning (RL) pipelines using autodifferentiation in MuJoCo-Warp to rapidly train humanoid musculoskeletal models for locomotion tasks.

National Aeronautics and Space Administration (NASA)

Launch Vehicle Flight Software Intern

Kennedy Space Center, FL (Jun 2026–Aug 2026)

- Developed a Python/pandas tool that automates the post-flight data review, building a source of truth for engineers to conduct health inspections on a commercial provider's launch vehicle software checks and HOOTL/HITL tests.
- Reviewed flight software pull requests on the customer's launch vehicle flight software, evaluating changes for safety-critical paths and downstream effects, and recommending code revisions and ticket reassignment.

Philips Cambridge NA HQ

Software Test Engineer Co-op

Cambridge, MA (Jan 2026–May 2026)

- Defined feature behavior for the PIC iX event notification system by writing Gherkin tests with the Solution Reliability team, producing 10+ Gherkin test cases for validation and verification.
- Built an internal RAG LLM within the Azure cloud pipeline to automate feature vulnerability collation, cutting manual defect analysis time by 60–70% and delivering real-time test coverage recommendations.

Synbio Technologies

Software Engineering Intern

South Brunswick, NJ (Jun 2024–Aug 2024)

- Developed a backend JavaScript event-logging system to track user interactions and auto-generate usage plots, enabling the team to pinpoint and resolve UX bottlenecks.
- Overhauled 100+ pages of the company website using behavior data from the logging system to prioritize redesign targets, boosting user engagement with interactive features post-launch.

PROJECTS

ORCA Hand Dexterous Manipulation — Physical AI Research Lead

Northeastern AI Club (Jun 2026–Present)

- Leading a student research group building on ETH Zurich's open-source ORCA Hand (17-DoF tendon-driven dexterous hand), constructed from scratch.
- Planning a contribution on soft-contact force dynamics for deformable objects, and building a clean data-collection pipeline to study zero-shot manipulation that reduces reliance on imitation learning.

Short Horizon Actor Critic Musculoskeletal Rapid Imitation Learning

Research Project (Mar 2026–Present)

- Implementing **deterministic** execution and contributing reverse-mode **autodifferentiation** support to MuJoCo-Warp, enabling end-to-end differentiable GPU simulation for RL, with the DeepMind MuJoCo and NVIDIA Warp teams.
- Testing SHAC RL policy training on MJWarp with cartpole and ant actors to later implement a rigid 22-muscle myoleg model for GPU-accelerated locomotion training.

Bipedal Spring Loaded Inverse Pendulum (bSLIP) Runner

Graduate Class Project (Sep 2025–Jan 2026)

- Implemented a two-leg bSLIP runner in MATLAB/Simulink (Simscape Multibody) with stance/swing PD control and contact/friction modeling.
- Performed apex-to-apex Poincare return-map analysis and added a one-step deadbeat regulator via touchdown hip-angle control to maintain apex height on rough terrain.

SKILLS AND HONORS

Languages

Python, Java, C, C++, MATLAB/Simulink, Verilog

Tools

Git, LTSpice, AWS, PyTorch, scikit-learn, Claude Code, GitHub Copilot, MuJoCo, CUDA, JAX, Isaac Sim, pandas, pytest, REST APIs, Azure, ROS2

Awards/Honors

1st in Simulation Division 2022 RCJ Thailand, Dean's List